

Imaadh Usoof

043-000-2805 | imaadh123@outlook.com | [linkedin.com/in/imaadh-usoof](https://www.linkedin.com/in/imaadh-usoof) | github.com/imaadhusoof

EDUCATION

University of Melbourne

Bachelor of Science, major in Computing and Software Systems

Melbourne, AUS

Jul. 2024 – Present

PROJECTS

Finance Portfolio Builder | *Python, FastAPI, React, TypeScript, AWS*

January 2026 – Present

- Built a full-stack investment-portfolio web app with a FastAPI backend and a React and TypeScript frontend, serving five years of daily price history and live-updating quotes
- Engineered a recommendation engine: a Markowitz mean-variance optimizer (SciPy SLSQP) with shrinkage estimators, mapping amount, risk tolerance and horizon to an allocation projected via Monte Carlo
- Deployed on AWS Lightsail behind an nginx reverse proxy with Let's Encrypt HTTPS; FastAPI runs as a systemd service with a nightly timer that refreshes the price cache- live at finance.imaadhusoof.com

Image Classification Pipeline | *Python, scikit-learn, PyTorch, ResNet50*

March 2026 – June 2026

- Engineered a 219-dimensional feature set and extracted 2048-dim ResNet50 embeddings benchmarking both against 5 classifiers (KNN, Logistic Regression, Linear SVM, RBF SVM, MLP) across coarse- and fine-grained classification tasks)
- Achieved 90.4 percent and 85.7 percent validation accuracy using ResNet50 embeddings, more than double the hand-crafted-feature baselines
- Designed and trained a 3-layer MLP (2048→512→128→10) with Adam optimization and cross-entropy loss, outperforming linear classifiers on the harder fine-grained task

Modular Tetris Game Engine | *Java, JUnit, object-oriented design*

March 2026 – June 2026

- Refactored an existing Tetris codebase into a modular, extensible architecture, applying GRASP principles and GoF design patterns (Decorator, Strategy, Builder)
- Decoupled core game logic, block behaviour, scoring, rendering, so new features could be added without touching existing code
- full UML documentation (domain model, class diagram, sequence diagrams) to drive the redesign before implementation
- Extended the redesigned engine with three new block-shape features, each covered by unit tests

Spellchecker Dictionary | *C, Patricia Trie* | Coursework

September 2025 – October 2025

- Implemented a spellchecker in C using a Patricia trie (compressed radix tree), supporting exact and fuzzy key lookup across a large address dataset
- Engineered bit-level string operations including custom getBit and createStem functions to navigate the trie using binary representations of characters
- Applied edit distance algorithms for closest-match recommendations when no exact key is found, traversing all descendant nodes at the mismatch point
- Conducted a formal complexity analysis comparing the Patricia trie against a linked list implementation, varying input size and key characteristics to validate Big-O behaviour empirically

PDF annotation app | *Electron, React, TypeScript, SQLite*

March 2025 – June 2025

- Built an application for lecture note taking and PDF annotation using Electron and TypeScript
- Implemented stylus and mouse-based annotation tools, used SQLite to store offline edits
- Ensured app has crucial PDF editing features only found on paid alternatives such as Adobe Acrobat

TECHNICAL SKILLS

Relevant Coursework: Probability, Algorithms and Data Structures, Machine Learning, Database Systems

Languages: Java, Python, C, SQL, JavaScript

Frameworks: FastAPI, React, Electron, SQLAlchemy

Databases: MySQL, PostgreSQL, Amazon RDS

Libraries: Tensorflow, pandas, NumPy, Matplotlib, pygame, PyTorch

Other Activities: Melbourne University Skiing Club, Melbourne University Golf Club, Boxing